

a) HLEDAŤME BAZIS B PODPROSTORU $\mathcal{L}(\{1, x, x^2\})$

$$u_1 = \frac{1}{\|1\|} \cdot 1 = \frac{1}{4}$$

$$u_2 = \int_1^5 \frac{x}{4} = 3$$

$$u_2 = \frac{1}{\|x-3\|} \cdot (x-3) = \frac{1}{\sqrt{\frac{16}{3}}} \cdot (x-3) = \frac{\sqrt{3}}{4} \cdot (x-3)$$

$$\boxed{x^2 - 6x + \frac{23}{3}}$$

$$u_3 = \left(\int_1^5 \frac{x^2}{4} \right) \cdot \frac{1}{4} + \left(\int_1^5 \frac{\sqrt{3}}{4} \cdot (x-3) \cdot x^2 \right) \cdot \frac{\sqrt{3}}{4} \cdot (x-3) = \frac{31}{12} + 72 \cdot (x-3)$$

$$u_3 = \frac{x^2 - 72x + \frac{2561}{12}}{\|x^2 - 72x + \frac{2561}{12}\|} = \sqrt{\frac{180}{4 \cdot 226 \cdot 629}} \cdot \left(x^2 - 72x + \frac{2561}{12} \right)$$

$$B = \left(\frac{1}{4}, \frac{\sqrt{3}}{4} \cdot (x-3), \sqrt{\frac{180}{4 \cdot 226 \cdot 629}} \cdot \left(x^2 - 72x + \frac{2561}{12} \right) \right)$$

b)

$$[\sin(x)]_V = \left\langle \sin(x), \frac{1}{4} \right\rangle + \left\langle \sin(x), \frac{\sqrt{3}}{4} \cdot (x-3) \right\rangle + \left\langle \sin(x), \sqrt{\frac{180}{4 \cdot 226 \cdot 629}} \cdot \left(x^2 - 72x + \frac{2561}{12} \right) \right\rangle =$$

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